Supplementary Material: Comprehensive SPARC MCMC Parameter Ledger

Brendon Boyd[1,*]
[1] *Independent Researcher, Burswood, Western Australia, Australia*
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I. INTRODUCTION

This supplementary document catalogues the comprehensive parameter estimation results for the entire 175-galaxy sample from the SPARC database [1]. Performed via parallelized Markov Chain Monte Carlo (MCMC) walkers (32 walkers × 3,000 steps per galaxy) within the Integrated Toroidal-Syntropic Model (ITSM) framework, these estimations resolve the stellar disk mass-to-light ratio ($\Upsilon_{\rm disk}$), bulge mass-to-light ratio ($\Upsilon_{\rm bulge}$), and the local Hubble expansion rate (H_0) tethered geometrically to the yield acceleration $a_0 = cH_0/2\pi$.

The ledger is categorized by galaxy name, optimized components, reduced chi-square (χ^2_ν), degrees of freedom (DoF), and quality categorization. Outliers that clipped the lower boundary wall ($H_0 \to 50.0$ km s^{-1}Mpc^{-1}) are designated as "Clipped" or "Standard" based on line-of-sight dust obscuration and turbulent compact kinematics, as analyzed in the main text [2].

II. METHODOLOGY: HIERARCHICAL BAYESIAN INFERENCE

The parameter extraction was executed using a hierarchical Markov Chain Monte Carlo (MCMC) ensemble sampler. The Integrated Toroidal-Syntropic Model (ITSM) imposes a rigid geometric dependency between the fundamental kinematic scale a_0 and the local Hubble parameter H_0, defined exactly by:

$$a_0 = \frac{cH_0}{2\pi} \tag{1}$$

where c is the speed of light. Consequently, the rotation curves are evaluated not as independent systems, but as nodes within a globally constrained Hubble flow. The kinematic profile of each galaxy is governed by the Plenum Shear Ansatz:

$$g_{\rm tot} = g_{\rm bar} + \frac{2}{3}\sqrt{g_{\rm bar} a_0} \tag{2}$$

where the pre-factor 2/3 derives strictly from the geometric projection of the transverse shear plane onto the bulk T^3 manifold.

A. MCMC Configuration and Priors

For each galaxy, we implement an affine-invariant ensemble sampler with 32 walkers initialized across a tight ball near the Maximum A Posteriori (MAP) estimate. Convergence is enforced over 3,000 steps per galaxy (with a 600-step burn-in discard phase). The log-posterior $\ln P(\theta|D)$ is defined by the product of the uniform priors and the Gaussian likelihood:

$$\ln \mathcal{L} = -0.5 \sum_i \left(\frac{V_{{\rm obs},i} - V_{{\rm ITSM},i}}{\sigma_{V,i}} \right)^2 \tag{3}$$

To ensure the physics remains scale-invariant, broad uniform priors are assigned to the baryonic mass-to-light ratios ($\Upsilon_{\rm disk} \in [0.01, 8.0]$, $\Upsilon_{\rm bulge} \in [0.00, 8.0]$). Convergence is verified per galaxy via Gelman-Rubin diagnostics and integrated autocorrelation times ($\tau \ll N_{\rm steps}$).

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* brendon.boyd@itsm-cosmology.org

III. COMPREHENSIVE SPARC PARAMETER LEDGER

Galaxy	$\Upsilon_{\rm disk}$	$\Upsilon_{\rm bulge}$	H_0	χ^2_ν	DoF	Quality
CamB	0.013	3.942	64.1	1.63	6	High-Fi
D512-2	0.278	3.754	71.9	2.27	1	High-Fi
D564-8	0.259	3.949	71.4	1.20	3	High-Fi
D631-7	0.114	4.237	71.6	1.63	13	High-Fi
DDO064	0.010	3.954	52.3	7.29	11	Standard
DDO154	0.349	4.027	72.1	4.70	9	High-Fi
DDO161	0.040	3.941	69.7	2.39	28	High-Fi
DDO168	0.140	4.040	71.4	1.52	7	High-Fi
DDO170	2.643	4.006	73.2	2.28	5	High-Fi
ESO079-G014	0.012	3.808	63.6	1.09	12	High-Fi
ESO116-G012	0.012	3.934	65.4	1.19	12	High-Fi
ESO444-G084	0.043	4.002	71.4	3.20	4	High-Fi
ESO563-G021	0.011	4.096	65.5	0.81	27	High-Fi
F561-1	0.164	3.797	71.2	0.95	3	High-Fi
F563-1	0.070	4.113	70.7	1.08	14	High-Fi
F563-V1	0.142	3.923	70.8	0.94	3	High-Fi
F563-V2	0.011	3.801	57.9	3.40	7	High-Fi
F565-V2	0.158	4.063	70.4	1.61	4	High-Fi
F567-2	0.227	3.992	71.8	2.36	2	High-Fi
F568-1	0.013	4.011	65.6	1.60	9	High-Fi
F568-3	0.015	3.960	65.7	0.66	15	High-Fi
F568-V1	0.012	3.961	59.3	1.71	12	High-Fi
F571-8	0.010	4.118	62.2	3.82	10	High-Fi
F571-V1	0.367	4.026	73.4	1.28	4	High-Fi
F574-1	0.015	3.937	66.7	0.76	11	High-Fi
F574-2	0.147	3.941	71.1	2.43	2	High-Fi
F579-V1	0.012	3.910	63.2	1.21	11	High-Fi
F583-1	0.011	4.006	55.1	2.05	22	High-Fi
F583-4	0.011	4.106	57.3	2.39	9	High-Fi
IC2574	0.056	3.948	71.7	0.40	31	High-Fi
IC4202	0.013	0.000	53.6	4.62	29	High-Fi
KK98-251	0.020	4.128	68.0	1.34	12	High-Fi
NGC0024	0.011	3.983	55.7	1.37	26	High-Fi
NGC0055	0.301	3.984	73.7	0.42	18	High-Fi
NGC0100	0.011	3.975	58.4	1.35	18	High-Fi
NGC0247	0.061	3.939	73.2	0.23	23	High-Fi
NGC0289	0.022	4.031	71.7	0.53	25	High-Fi
NGC0300	0.045	4.178	71.2	0.33	22	High-Fi
NGC0801	0.013	4.032	73.4	0.84	10	High-Fi
NGC0891	0.012	0.010	68.8	1.40	15	High-Fi
NGC1003	0.066	3.887	72.8	0.56	33	High-Fi
NGC1090	0.011	4.173	64.9	1.10	21	High-Fi
NGC1705	0.013	4.114	68.3	1.02	11	High-Fi
NGC2366	0.011	4.051	56.9	2.82	23	High-Fi
NGC2403	0.010	4.077	57.1	1.14	70	High-Fi
NGC2683	0.099	4.113	72.3	0.48	8	High-Fi
NGC2841	0.026	4.006	63.0	0.65	47	High-Fi
NGC2903	0.011	3.860	69.6	0.87	31	High-Fi
NGC2915	0.021	3.887	70.0	0.72	27	High-Fi
NGC2955	0.011	0.000	59.9	2.53	21	High-Fi
NGC2976	0.010	3.956	52.8	3.14	24	High-Fi
NGC2998	0.010	4.002	58.6	4.58	10	High-Fi
NGC3109	0.018	4.106	68.4	1.27	22	High-Fi
NGC3198	0.011	3.926	62.8	0.57	40	High-Fi
NGC3521	0.011	4.056	65.1	0.83	38	High-Fi
NGC3726	0.128	4.045	72.8	0.32	9	High-Fi
NGC3741	0.047	4.076	71.1	4.66	18	High-Fi
NGC3769	0.128	4.197	74.3	0.44	9	High-Fi
NGC3877	0.018	3.945	72.7	0.42	10	High-Fi

NGC3893	0.052	3.998	75.2	0.34	7	High-Fi
NGC3917	0.022	3.933	72.3	0.33	14	High-Fi
NGC3949	0.043	3.924	73.9	0.53	4	High-Fi
NGC3953	0.104	3.784	72.9	0.40	5	High-Fi
NGC3972	0.028	4.026	73.1	0.52	7	High-Fi
NGC3992	0.931	3.745	74.9	0.34	6	High-Fi
NGC4010	0.018	3.893	70.3	0.60	9	High-Fi
NGC4013	0.286	3.945	65.1	0.35	33	High-Fi
NGC4051	0.076	3.929	73.2	0.57	4	High-Fi
NGC4068	0.015	3.984	66.2	3.13	3	High-Fi
NGC4085	0.019	4.127	72.4	0.94	4	High-Fi
NGC4088	0.029	4.050	73.1	0.37	9	High-Fi
NGC4100	0.053	4.028	73.3	0.15	21	High-Fi
NGC4138	0.202	4.067	71.5	1.32	4	High-Fi
NGC4157	0.021	4.042	73.3	0.27	14	High-Fi
NGC4183	0.034	4.201	70.8	0.39	20	High-Fi
NGC4214	0.013	4.181	68.5	0.91	11	High-Fi
NGC4217	0.027	0.000	51.7	14.75	16	Standard
NGC4389	0.019	4.145	73.4	1.20	3	High-Fi
NGC4559	0.017	4.092	70.4	0.41	29	High-Fi
NGC5005	0.010	0.000	56.8	4.00	15	High-Fi
NGC5033	0.012	0.001	64.6	1.50	19	High-Fi
NGC5055	0.012	4.030	71.0	0.42	25	High-Fi
NGC5371	0.027	4.060	72.4	0.21	16	High-Fi
NGC5585	0.010	3.927	56.3	2.84	21	High-Fi
NGC5907	0.173	3.848	73.7	0.22	16	High-Fi
NGC5985	0.036	4.176	68.0	0.42	30	High-Fi
NGC6015	0.011	4.167	66.9	0.56	41	High-Fi
NGC6195	0.016	0.000	57.4	3.87	20	High-Fi
NGC6503	0.015	3.955	71.2	0.33	28	High-Fi
NGC6674	0.036	3.840	68.9	0.75	12	High-Fi
NGC6789	0.011	4.080	60.1	17.13	1	Standard
NGC6946	0.010	0.000	52.5	3.88	55	High-Fi
NGC7331	0.021	4.040	71.4	0.16	33	High-Fi
NGC7793	0.010	4.034	52.1	3.19	43	High-Fi
NGC7814	0.016	0.000	52.9	11.75	15	Standard
PGC51017	0.029	4.209	70.1	1.82	3	High-Fi
UGC00128	0.051	4.123	70.7	0.95	19	High-Fi
UGC00191	0.025	3.880	70.4	0.92	6	High-Fi
UGC00634	3.788	3.853	74.6	3.28	1	High-Fi
UGC00731	0.328	4.026	72.2	3.91	9	High-Fi
UGC00891	1.916	3.916	72.3	2.66	2	High-Fi
UGC01230	0.026	3.884	69.6	1.24	8	High-Fi
UGC01281	0.010	4.024	51.1	9.55	22	Standard
UGC02023	0.030	3.887	70.5	2.10	2	High-Fi
UGC02259	0.112	4.083	72.5	0.77	5	High-Fi
UGC02455	0.018	3.807	72.4	0.92	5	High-Fi
UGC02487	0.277	3.910	69.7	0.65	14	High-Fi
UGC02885	0.020	0.008	66.9	1.42	16	High-Fi
UGC02916	0.011	0.000	50.3	24.83	40	Clipped
UGC02953	0.010	0.000	50.2	16.82	112	Clipped
UGC03205	0.010	0.000	50.9	7.32	45	Clipped
UGC03546	0.011	0.000	52.7	5.58	27	Standard
UGC03580	0.010	0.000	50.6	8.68	44	Clipped
UGC04278	0.011	4.137	53.4	2.48	22	High-Fi
UGC04305	0.012	4.010	59.6	1.35	19	High-Fi
UGC04325	0.037	4.052	70.6	0.79	5	High-Fi
UGC04483	0.012	3.817	60.0	4.75	5	High-Fi
UGC04499	0.137	4.179	71.9	0.93	6	High-Fi
UGC05005	0.028	4.019	69.2	1.29	8	High-Fi
UGC05253	0.010	0.000	50.2	30.12	70	Clipped
UGC05414	0.054	3.982	71.9	1.11	3	High-Fi
UGC05716	0.288	3.923	71.3	2.31	9	High-Fi
UGC05721	0.011	4.045	57.9	2.48	20	High-Fi

UGC05750	0.012	3.818	60.2	2.07	8	High-Fi
UGC05764	0.050	4.039	71.9	2.57	7	High-Fi
UGC05829	0.025	3.999	67.6	1.96	8	High-Fi
UGC05918	0.041	4.150	69.3	0.93	5	High-Fi
UGC05986	0.023	3.918	70.1	0.50	12	High-Fi
UGC05999	2.619	4.033	72.7	2.03	2	High-Fi
UGC06399	0.040	4.020	70.6	0.68	6	High-Fi
UGC06446	0.039	3.892	71.4	1.20	14	High-Fi
UGC06614	0.177	0.011	53.6	11.17	10	Standard
UGC06628	0.070	4.072	70.9	0.87	4	High-Fi
UGC06667	0.049	4.009	71.8	1.33	6	High-Fi
UGC06786	0.010	0.000	50.2	30.03	42	Clipped
UGC06787	0.010	0.000	50.1	46.07	68	Clipped
UGC06818	0.060	3.902	71.2	0.67	5	High-Fi
UGC06917	0.199	4.057	73.4	0.37	8	High-Fi
UGC06923	0.099	3.880	72.4	0.81	3	High-Fi
UGC06930	0.171	4.023	73.1	0.62	7	High-Fi
UGC06973	0.036	4.087	73.2	0.62	6	High-Fi
UGC06983	0.196	3.851	74.0	0.58	14	High-Fi
UGC07089	0.039	3.939	70.2	0.45	9	High-Fi
UGC07125	0.222	3.994	72.6	1.10	10	High-Fi
UGC07151	0.021	3.930	70.6	0.68	8	High-Fi
UGC07232	0.016	3.890	67.8	5.66	1	Standard
UGC07261	0.047	3.878	70.9	1.04	4	High-Fi
UGC07323	0.019	3.829	69.6	0.75	7	High-Fi
UGC07399	0.055	4.143	73.1	0.61	7	High-Fi
UGC07524	0.022	3.849	69.0	0.46	28	High-Fi
UGC07559	0.030	3.817	70.2	1.33	4	High-Fi
UGC07577	0.013	4.070	60.8	2.98	6	High-Fi
UGC07603	0.027	4.094	71.3	0.70	9	High-Fi
UGC07608	0.064	4.053	72.0	1.31	5	High-Fi
UGC07690	0.058	4.123	73.6	0.87	4	High-Fi
UGC07866	0.023	3.997	69.2	1.31	4	High-Fi
UGC08286	0.021	4.019	70.7	0.74	14	High-Fi
UGC08490	0.018	3.962	71.8	0.78	27	High-Fi
UGC08550	0.051	3.675	71.7	1.13	8	High-Fi
UGC08699	0.010	0.000	51.2	6.90	38	Standard
UGC08837	0.033	4.020	70.0	1.37	5	High-Fi
UGC09037	0.104	4.054	73.4	0.23	19	High-Fi
UGC09133	0.010	0.000	51.3	5.18	65	Standard
UGC09992	0.130	4.055	71.6	1.42	2	High-Fi
UGC10310	0.075	4.019	70.1	1.03	4	High-Fi
UGC11455	0.011	3.752	66.8	0.55	33	High-Fi
UGC11557	0.012	3.905	64.6	1.49	9	High-Fi
UGC11820	0.012	4.067	61.0	3.98	7	High-Fi
UGC11914	0.010	0.000	51.1	5.98	62	Standard
UGC12506	0.013	3.861	67.1	0.48	28	High-Fi
UGC12632	0.040	3.911	69.6	0.82	12	High-Fi
UGC12732	0.089	3.898	72.0	1.36	13	High-Fi
UGCA281	0.011	4.065	54.8	8.94	4	Standard
UGCA442	0.086	3.956	71.6	3.07	5	High-Fi
UGCA444	0.132	3.963	72.5	5.18	33	Standard

[1] F. Lelli, S. S. McGaugh, and J. M. Schombert, The Astronomical Journal **152**, 157 (2016).
[2] B. Boyd, The integrated toroidal-syntropic model: Relativistic field equations, topology-induced superfluid dynamics, and multi-scale falsifiability (2026), main Manuscript.